

Programming Exam 7

Date/Time: 2026.05.26 08:00 – 08:50

(程式檔命名學號_pe7.py，上傳至 Moodle PE7 上傳區)

Problem: Forum Reply Analysis

A variable forum dictionary (in **forum1.txt** and **forum2.txt**) represents posts and their nested reply structure. Each comment dictionary maps a comment_id (int) to another dictionary, as shown below.

```
forum = {
    post_id (int): {
        'title': str, # post title
        'comments': dict # first-level replies
    },
    ...
}
```

value in comment_id dictionary:

```
{
    'author': str, # commenter name
    'likes': int, # this comment's like count
    'replies': dict # nested replies (same structure, unlimited depth)
}
```

Your goal is to *create some functions by yourself* that can help solve the problem. Then combine your created functions in an overall `analyze_forum` function, which prints out an outcome dictionary with the following requirements and structure.

- **total_likes**: A dictionary mapping {post_id: total_likes,...} where total_likes is the sum of **all** likes in **every** reply at **any** depth under that post.
- **max_likes_post**: The single post_id whose total_likes is highest.
- **flattened**: A list of tuples (post_id, comment_id, depth, likes), where depth = 1 for first-level comments, 2 for their replies, 3 for replies to replies, etc.
- **author_likes**: A dictionary mapping {author: total_likes,...} summing each author's likes **across all posts and all depths**.

```
# Structure of the output dictionary returned by analyze_forum()
output = {
    'total_likes': {
        # post_id (int) : sum of likes across all nested comments for that post
        post_id: total_likes,
        ...
    },
    'max_likes_post': post_id,
    # the single post_id whose total_likes value is the highest

    'flattened': [
        # a list of tuples, one per comment at any depth:
        # (post_id, comment_id, depth, likes)
        (post_id, comment_id, depth, likes),
        ...
    ],

    'author_likes': {
        # commenter_name (str) : sum of that author's likes across all posts/depths
        author_name: total_likes,
        ...
    }
}
```

重要規定：

- (1) 從此次 PE7 開始，允許同學們使用生成式 AI 工具如 ChatGPT 來協助作答。
- (2) 請務必使用自己能夠理解的 **Python 程式語法** 來作答。

若我們認為你使用超過考試範圍內教的語法以外的進階或特殊語法，將會請你前來說明解釋該語法如何運作。若無法清楚說明該語法如何輔助解出此題，將視狀況對分數打折扣或不予計分。

Sample Input/Output

Sample 1	Sample 2
<u>Input (forum1.txt)</u> <pre> forum = { 101: { 'title': 'Python', 'comments': { 1001: { 'author': 'Alice', 'likes': 10, 'replies': { 10011: { 'author': 'Bob', 'likes': 5, 'replies': { 100111: { 'author': 'Alice', 'likes': 3, 'replies': {} } } } } }, 1002: { 'author': 'Charlie', 'likes': 2, 'replies': {} } } }, 102: { 'title': 'Statistics', 'comments': {} } } </pre> <u>Output</u> (程式執行後須印出的結果) Need to be “exactly” the same as the output. <pre> D:\Dropbox\計算機概論2025Spring\PE>python pe7.py { 'total_likes': { 101: 20, 102: 0, }, 'max_likes_post': 101, 'flattened': [(101, 1001, 1, 10), (101, 10011, 2, 5), (101, 100111, 3, 3), (101, 1002, 1, 2)], 'author_likes': { 'Alice': 13, 'Bob': 5, 'Charlie': 2, } } D:\Dropbox\計算機概論2025Spring\PE> </pre>	<u>Input (forum2.txt)</u> <u>Output</u> (程式執行後須印出的結果) Need to be “exactly” the same as the output. <pre> D:\Dropbox\計算機概論2025Spring\PE>python pe7.py { 'total_likes': { 1: 402, 2: 149, 3: 233, 4: 658, }, 'max_likes_post': 4, 'flattened': [(1, 1, 1, 38), (1, 2, 2, 75), (1, 3, 3, 58), (1, 4, 4, 57), (1, 5, 1, 33), (1, 6, 1, 23), (1, 7, 1, 23), (1, 8, 1, 95), (2, 9, 1, 33), (2, 10, 2, 13), (2, 11, 3, 68), (2, 12, 1, 0), (2, 13, 1, 14), (2, 14, 1, 21), (3, 15, 1, 61), (3, 16, 1, 88), (3, 17, 1, 84), (4, 18, 1, 77), (4, 19, 2, 62), (4, 20, 3, 79), (4, 21, 4, 46), (4, 22, 5, 86), (4, 23, 1, 86), (4, 24, 1, 38), (4, 25, 1, 16), (4, 26, 1, 1), (4, 27, 1, 14), (4, 28, 1, 89), (4, 29, 1, 3), (4, 30, 1, 61)], 'author_likes': { 'Leo': 407, 'Frank': 98, 'Karl': 131, 'Judy': 112, 'Diana': 101, 'Bob': 111, 'Grace': 204, 'Alice': 110, 'Heidi': 88, 'Eve': 79, 'Charlie': 1, } } D:\Dropbox\計算機概論2025Spring\PE> </pre>

(繳交是交 pe7.py 檔，不是交截圖)

Note: You need to write comments (註解) for each part in your code.